of configuration options available like block storage, hypervisors, databases, message queuing, networking, object storage, source builds, etc. It works on agent-based architecture, where every VM machine is controlled by a central master agent.

The most important deployment scenarios for Chef are: All-in-one-compute, single controller + N compute and Vagrant. Chef manages the entire infrastructure by converting it into code, and provides configuration management and other infrastructure management tasks using Recipes.

Components

Chef has the following components.

- *Chef Cookbooks:* These are part of the configuration and policy distribution, and comprise code that describes everything about the infrastructure.
- *Chef Server:* This acts as the central repository for Cookbooks and every node attached to the entire infrastructure.
- *Chef Client:* This runs on every node attached in the cloud and communicates with the Chef Server to replicate the latest configuration description.
- *Chef DK:* This provides developers with various tools to develop infrastructure automation code.

Figure 1 highlights all the components of the Chef Automation Framework tool. Figure 2 illustrates how the Chef automation tool works.

Features

The features of Chef include the following:

- It allows the management of servers ranging from five to 50,000 by converting the infrastructure into code. It makes the infrastructure highly flexible, versionable and properly testable.
- It provides many resources for configuring SaaS services, and integrates cloud provisioning APIs as well as third party software like HashiCorp's Terraform.
- It includes the Chef Development Kit (ChefDK), which has robust testing tools for validating infrastructure changes before implementing them in real-time.
- Highly customised software provides support for unique and complex infrastructure environments.
- Built-in validations ensure that configurations are changed if a system diverges from the desired state, as per the user.

Official website: *https://www.chef.io/chef/*
Latest versions: Chef Workstation: 0.1.162; Chef Server: 12.17.33; Chef Client: 14.5.33; Chef Development Kit: 3.3.23

Ansible

Ansible is another open source infrastructure automation tool for OpenStack. It provides support to configure systems, deploy software and orchestrate more advanced IT-intensive tasks like consistent deployments and zero downtime while installing updates. Ansible goes beyond more than simple deployment. After deploying OpenStack, Ansible OpenStack modules can be used to manage all sorts of cloud operations. Ansible provides powerful tools for deploying and managing OpenStack — to provision, configure and deploy applications and services on top of the cloud.

OpenStack-Ansible is an official OpenStack project that aims to deploy production environments from source in a manner that provides high scalability for systems administrators to operate, upgrade and transform the OpenStack installation. It is based on an agentless architecture, so there is no requirement to configure VMs or workstations prior to installation. Ansible can work directly with them through the command line.

Ansible provides playbooks and roles for performing various deployment and configuration tasks in the OpenStack cloud. It provides systems administrators with a control language that uses modules or routines to perform all sorts of tasks on nodes, which users can control through the command line.

Figure 1: Chef components

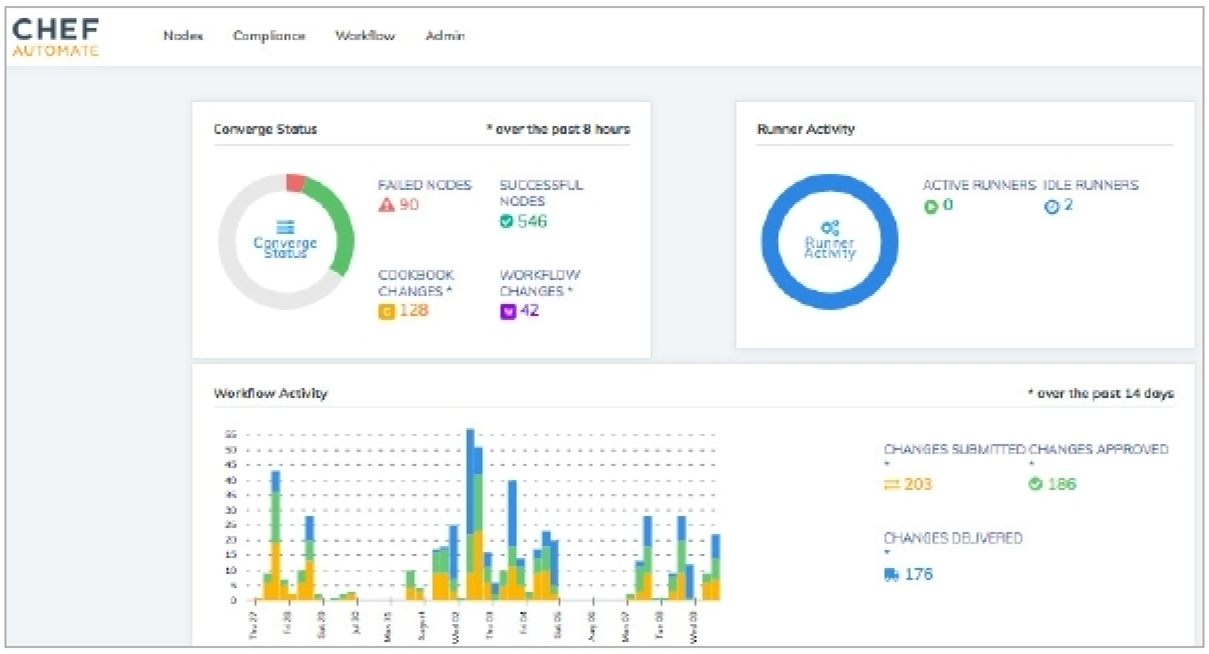

Figure 2: Chef Automate GUI interface